

ASHGABAT, Turkmenistan

WMO: 388800

Lat: 37.987N

Lon: 58.361E

Elev: 211

StdP: 98.82

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.8	-4.8	-12.2	1.3	-2.3	-9.8	1.7	1.7	9.1	7.1	7.7	6.9	2.5	100	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	40.9	19.9	39.4	19.5	38.1	19.3	23.4	35.0	22.4	34.2	21.5	33.5	4.3	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.3	30.5	18.1	13.4	29.8	17.0	12.5	28.7	70.7	34.9	66.9	34.3	63.6	33.4	33.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.1	7.1	-9.9	43.3	4.0	1.4	-12.8	44.3	-15.1	45.2	-17.4	46.0	-20.3	47.0		
(5)			-10.4	25.2	4.0	2.0	-13.3	26.6	-15.6	27.8	-17.9	28.9	-20.8	30.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.7	4.2	5.9	11.4	17.6	24.2	29.2	31.5	30.1	24.5	17.6	10.3	5.4	(6)
(7)		DBStd	10.54	4.59	5.14	4.80	4.82	3.53	2.85	2.43	2.70	3.49	4.46	4.74	4.52	(7)
(8)		HDD10.0	571	187	131	39	4	0	0	0	0	0	3	52	155	(8)
(9)		HDD18.3	1794	438	349	220	68	3	0	0	0	2	66	245	401	(9)
(10)		CDD10.0	3389	7	14	83	233	440	577	665	622	434	240	60	13	(10)
(11)		CDD18.3	1570	0	0	6	48	185	327	407	364	186	44	3	1	(11)
(12)		CDH23.3	22058	0	5	110	522	2116	4752	6372	5432	2221	486	38	4	(12)
(13)		CDH26.7	12859	0	1	37	203	1008	2837	4112	3394	1089	167	10	1	(13)
(14)	Wind	WSAvg	3.1	2.6	2.9	3.2	3.4	3.5	3.7	3.7	3.3	3.0	2.7	2.4	2.5	(14)
(15)	Precipitation	PrecAvg	234	22	31	40	40	28	7	3	1	3	13	21	20	(15)
(16)		PrecMax	398	72	83	100	118	84	43	27	11	31	40	70	55	(16)
(17)		PrecMin	142	0	1	14	1	1	0	0	0	0	0	0	0	(17)
(18)		PrecStd	59	17	20	18	30	22	9	6	2	6	13	17	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.9	23.7	31.0	35.0	38.1	41.8	43.0	42.2	38.2	34.8	27.7	23.1	(19)
(20)			MCWB	10.6	11.8	15.2	17.5	19.0	19.1	20.9	20.1	18.1	17.0	14.7	11.8	(20)
(21)		2%	DB	17.1	19.8	26.2	31.8	35.8	39.8	41.1	40.2	36.0	30.8	23.1	18.8	(21)
(22)			MCWB	9.0	10.2	13.7	16.7	18.3	19.3	20.3	19.4	17.8	16.3	13.1	10.0	(22)
(23)		5%	DB	13.4	16.2	23.0	28.9	33.9	38.2	39.9	38.9	34.1	28.1	20.1	14.9	(23)
(24)			MCWB	7.8	9.1	12.9	16.0	17.8	18.8	19.9	19.2	17.9	15.5	12.2	8.8	(24)
(25)		10%	DB	10.7	13.4	19.8	26.1	32.0	36.9	38.2	37.4	32.2	25.9	17.8	11.9	(25)
(26)			MCWB	6.6	8.1	12.0	15.1	17.4	18.7	19.9	19.1	17.3	14.5	11.4	7.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	12.8	16.3	19.7	21.9	22.6	24.7	24.2	22.0	19.2	16.1	12.3	(27)
(28)			MADB	18.3	20.5	26.9	29.5	31.0	34.9	36.4	36.0	33.1	28.0	23.4	20.4	(28)
(29)		2%	WB	9.5	11.0	14.7	18.2	20.0	21.6	23.6	23.2	20.7	17.7	14.5	10.6	(29)
(30)			MADB	15.8	17.6	23.6	27.8	30.8	34.5	35.6	34.6	31.0	27.1	20.7	16.9	(30)
(31)		5%	WB	8.2	9.8	13.5	17.1	19.2	20.8	22.8	22.2	19.7	16.7	13.2	9.2	(31)
(32)			MADB	12.9	15.1	21.2	26.0	29.9	33.7	34.7	33.5	30.0	25.4	18.4	14.0	(32)
(33)		10%	WB	6.9	8.6	12.3	16.1	18.5	20.1	21.9	21.0	18.7	15.6	11.9	8.0	(33)
(34)			MADB	10.2	12.7	18.8	24.2	28.7	32.6	34.1	32.6	29.2	24.0	17.0	11.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.8	9.6	10.5	11.2	12.2	12.9	12.8	13.7	13.7	12.6	10.3	8.8	(35)
(36)			MCDBR	14.3	15.5	16.8	15.8	15.3	15.5	14.7	15.2	15.7	15.5	14.7	14.6	(36)
(37)			MCWBR	8.0	8.2	7.2	5.8	5.3	5.4	5.2	5.7	6.0	6.4	7.1	8.0	(37)
(38)		5% WB	MADB	12.7	13.4	14.6	13.0	12.6	12.2	11.2	11.7	12.0	13.2	12.2	12.5	(38)
(39)			MCWBR	7.4	7.5	6.9	5.5	4.9	4.8	5.2	5.7	5.8	6.0	6.3	7.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.419	0.478	0.571	0.632	0.634	0.624	0.630	0.570	0.530	0.548	0.481	0.424	(40)	
(41)		taud	2.006	1.860	1.690	1.622	1.649	1.669	1.662	1.762	1.818	1.774	1.893	1.993	(41)	
(42)		Ebn at Noon	714	718	690	680	691	698	689	719	717	640	635	671	(42)	
(43)		Edh at Noon	130	171	223	253	250	245	245	217	193	184	145	123	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.19	3.00	4.14	5.45	6.71	7.87	7.74	7.20	5.90	4.10	2.69	1.99	(44)	
(45)		RadStd	0.23	0.34	0.32	0.55	0.33	0.24	0.23	0.29	0.17	0.24	0.23	0.26	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.41	N/A	N/A	+0.78	N/A	N/A	N/A	N/A	+112	+101
(47) Regional (4 neighbors)		N/A	N/A	N/A	+0.64	N/A	N/A	N/A	N/A	N/A	+82

Nomenclature: See separate page